UAVSAR ΔSWE Retrievals at Mores Creek Summit:

Selected InSAR pairs for sub-km precipitation-pattern input to snow models

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1 Purpose and data context

Snow models are usually forced with precipitation fields far coarser than the sub-km accumulation patterns that terrain imposes. L-band repeat-pass InSAR measures the phase change caused by SWE gained between two flights, at meter-scale resolution. This report summarizes a pilot at Mores Creek Summit (MCS), Idaho, that turns 22 SnowEx UAVSAR interferometric pairs (winters 2019–20 and 2020–21, lines 05208 and 23205) into ΔSWE maps, evaluates them against airborne lidar snow depth and the in-domain SNOTEL, and selects the pairs most useful for defining the sub-km precipitation pattern. Figure 1 places every UAVSAR flight and both QSI lidar surveys on the SNOTEL SWE record.

2 Recommended pairs

Table 1 lists the five ranked pairs plus one additional example. The rank criterion is the mean (over four polarizations) Pearson correlation at 90 m between the pair's ΔSWE map and the season-matched lidar snow depth, restricted to pairs whose interval contains real snowfall (SNOTEL ΔSWE $\geq +10$ mm) with adequate valid area ($n_{90} \geq 500$ cells). Pair 4 fails only the snowfall gate: a near-zero-change interval whose retrieval still tracks the snowpack pattern ($r = +0.49$ against the lidar it brackets), making it a clean null/redistribution test case for a model.

Table 1: Top 5 pairs for precipitation-pattern input, plus the zero-change example. r vs lidar is the polarization-mean correlation with the season-matched QSI snow depth at 90 m; cross-pol is the mean inter-polarization correlation of the retrieval.

rank	pair	line	dates	SNOTEL ΔSWE (mm)	r vs lidar	cross-pol	n_{90}
1	8	05208	2021-02-10 → 2021-03-03	+173	+0.637	+0.89	3220
2	2	23205	2021-03-10 → 2021-03-22	+28	+0.577	+0.95	3685
3	1	23205	2021-03-16 → 2021-03-22	+28	+0.483	+0.88	3594
4	5	23205	2021-03-03 → 2021-03-16	+13	+0.466	+0.90	3729
5	6	05208	2021-03-03 → 2021-03-10	+13	+0.454	+0.94	4413
–	4	23205	2021-03-10 → 2021-03-16	+0	+0.486	+0.95	3819

Two features strengthen the top choices: pair 8 correlates nearly equally with the 2020 lidar (+0.650) and the 2021 lidar (+0.637), evidence that the accumulation pattern it captures is the persistent, terrain-controlled one a model precipitation field should inherit; and pairs 5/6 sample

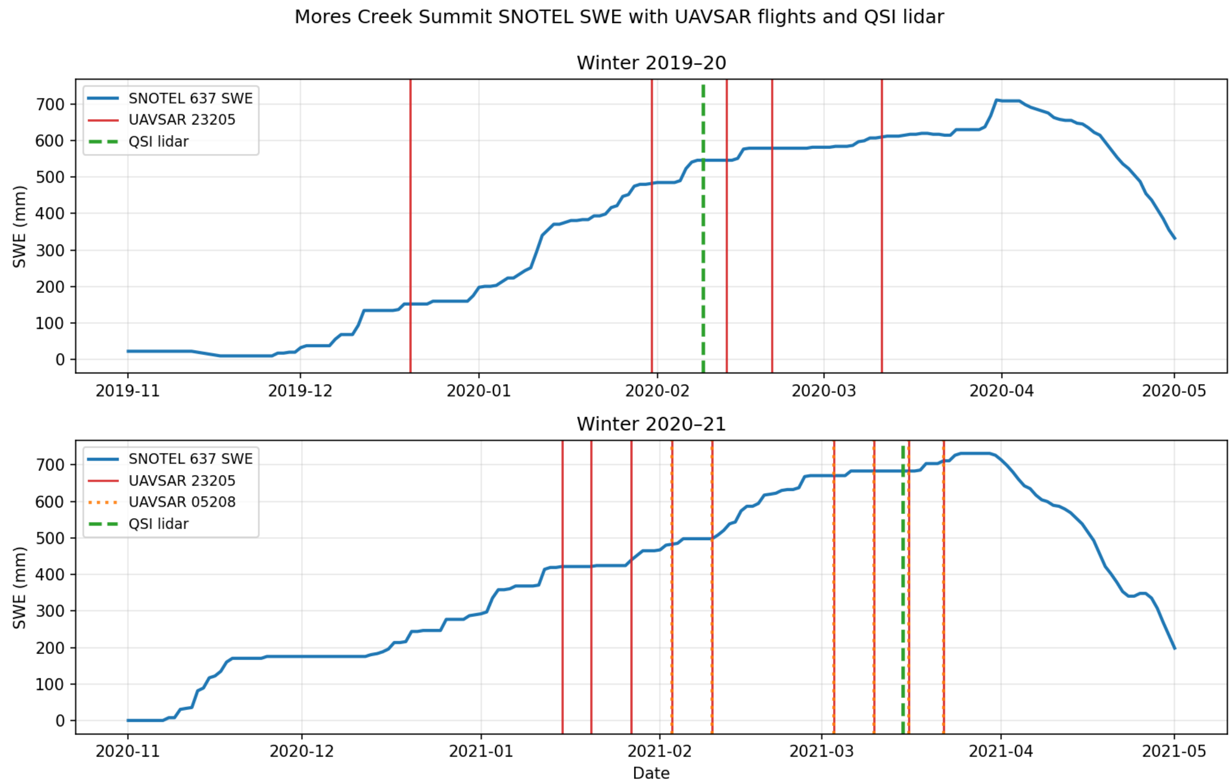

Figure 1: Mores Creek Summit SNOTEL (637:ID:SNTL) daily SWE for the two campaign winters. Vertical lines: UAVSAR flights (line 23205 solid red, line 05208 dotted orange; the two lines flew the same days in 2021). Dashed green: QSI lidar snow-depth surveys (2020-02-09 and 2021-03-15).

the same early-March snowfall from opposite headings, giving a built-in cross-view consistency check.

3 Processing steps

3.1 Site cropping and the canonical grid

All products are cropped to the MCS lidar footprint and expressed on one canonical grid: 3 m, UTM 11N (EPSG:32611), 2801 × 2624 cells, derived from the native 0.5 m QSI lidar grid. Everything downstream — lidar, phase, coherence, incidence angle, atmosphere, retrievals — shares this grid, so no resampling is needed at analysis time.

3.2 Regridding to 3 m

Lidar products (snow depth, DTM, DSM, CHM) are block-averaged 0.5 m → 3 m. UAVSAR ground-projected products are reprojected from their native equiangular grid: coherence and unwrapped phase bilinearly, and *wrapped* phase in the complex domain (averaging real and imaginary parts separately) so that phase wrapping is respected.

3.3 Atmospheric correction

For each of the 20 flights, ERA5 pressure-level reanalysis at the acquisition hour gives a refractivity profile from which the two-way phase delay is integrated *below the aircraft altitude* ($\sim$12.5 km — UAVSAR flies inside the atmosphere, so the standard satellite formulation is referenced to the aircraft, not the model top), mapped through the local incidence angle. Per-pair differences (`atm_delay_diff`) are subtracted from the unwrapped phase.

3.4 Local incidence angle

Per-line local incidence angle comes from the JPL Stack look vectors (.lkv/.llh files) combined with the 2023-02-09 lidar DTM, mosaicked across the three along-track segments and geocoded directly onto the canonical grid. Grazing and back-facing terrain ($\cos(\text{LIA}) \leq 0.05$) is masked.

3.5 Unwrapping of pairs without an ASF product

Five pairs (9, 12, 14, 20, 21) ship no unwrapped phase from ASF. Their wrapped interferograms were multilooked 5×5 (to 15 m), unwrapped with SNAPHU (smooth-cost, MCF initialization, coherence-weighted), and restored to 3 m by choosing, at every pixel, the 2π branch of the original wrapped phase nearest the upsampled unwrapped field — exact rewrap consistency by construction. Caveat: disconnected unwrapped components carry independent phase constants (most relevant for the low-coherence 2020 pairs 20 and 21).

3.6 Phase ramp removal

Pair 8 (05208, 2021-02-10 → 03-03, the 173 mm storm sequence) carried a residual long-wavelength tilt ($\sim$0.15 rad km^{-1}), typical of small aircraft-motion / baseline errors. The correction fits $\varphi = a + bx + cy + d\,z(\text{DEM})$ at 90 m and removes only the planar part; the elevation-correlated term (+0.34 rad per 100 m) is deliberately *kept*, since orographic accumulation is itself elevation-correlated. The pair's lidar correlation improved from +0.635 to +0.672.

3.7 ΔSWE inversion and validation

Corrected phase is inverted with the Guneriussen et al. (2001) dry-snow relation using the real per-pixel incidence angle and new-snow density 250 kg m^{-3}, masked at coherence < 0.35, and anchored so the 90 m window at the MCS SNOTEL matches the station's interval ΔSWE (SNOTEL is therefore a *reference*, not an independent check). Four validations: (i) phase-closure triangles, RMS 0.5–1.6 rad $\approx$ 7–21 mm w.e. for ASF-product triangles (24–31 mm where a custom-unwrapped pair participates) — small against 50–170 mm storm signals; (ii) cross-polarization consistency $r = 0.71$–0.96 for the well-observed pairs; (iii) SNOTEL anchor diagnostics (offsets and interval densities logged per pair); (iv) lidar comparisons — the 2021-03-15 survey falls inside pair 4's interval ($r = +0.51$ HH), and the top-ranked pairs reach $r = +0.45$ to $+0.65$.

4 Deliverables

- `mcs_top_pairs_subset.nc` (1.2 GB NetCDF4): the six pairs of Table 1 in rank order (`rank_note` coordinate), with ΔSWE, coherence, unwrapped phase and atmospheric correction per pair, plus the two QSI snow-depth surveys, the DEM, and both lines' incidence angle. Format details in the companion user guide.

- `mcs_zarr_userguide.pdf`: complete format specification of the master datacube and of the subset file.

A Full pair ranking

Table 2: All 22 pairs: polarization-mean correlation of ΔSWE with each lidar survey at 90 m, SNOTEL interval ΔSWE (mm), valid 90 m cells, cross-polarization consistency, and gate/provenance notes. Ranking uses the season-matched lidar column.

pair	line	dates	SNOTEL	r 2020	r 2021	n_{90}	cross-pol	notes
0	05208	2021-03-16 → 2021-03-22	+28	+0.061	+0.185	3157	+0.82	rank 11
1	23205	2021-03-16 → 2021-03-22	+28	+0.229	+0.483	3594	+0.88	rank 3
2	23205	2021-03-10 → 2021-03-22	+28	+0.362	+0.577	3685	+0.95	rank 2
3	05208	2021-03-10 → 2021-03-16	+0	-0.210	-0.292	4415	+0.95	no snowfall
4	23205	2021-03-10 → 2021-03-16	+0	+0.370	+0.486	3819	+0.95	zero-change example, no snowfall
5	23205	2021-03-03 → 2021-03-16	+13	+0.238	+0.466	3729	+0.90	rank 4
6	05208	2021-03-03 → 2021-03-10	+13	+0.335	+0.454	4413	+0.94	rank 5
7	23205	2021-03-03 → 2021-03-10	+13	-0.028	+0.170	3679	+0.91	rank 12
8	05208	2021-02-10 → 2021-03-03	+173	+0.650	+0.637	3220	+0.89	rank 1
9	23205	2021-02-10 → 2021-03-03	+173	+0.257	+0.318	3723	+0.87	rank 6, SNAPHU unwrap
10	05208	2021-02-03 → 2021-02-10	+15	+0.123	-0.142	4394	+0.96	rank 18
11	23205	2021-02-03 → 2021-02-10	+15	+0.187	+0.109	3157	+0.94	rank 14
12	23205	2021-01-27 → 2021-02-10	+58	+0.289	+0.140	3408	+0.41	rank 13, SNAPHU unwrap
13	23205	2021-01-27 → 2021-02-03	+43	+0.251	+0.255	653	+0.89	rank 8
14	23205	2021-01-20 → 2021-02-03	+61	-0.027	-0.046	3191	+0.67	rank 16, SNAPHU unwrap
15	23205	2021-01-20 → 2021-01-27	+18	-0.050	+0.002	3727	+0.72	rank 15
16	23205	2021-01-15 → 2021-01-27	+18	-0.402	-0.467	3209	+0.92	rank 19
17	23205	2021-01-15 → 2021-01-20	+0	-0.309	-0.434	3695	+0.82	no snowfall
18	23205	2020-02-21 → 2020-03-11	+30	+0.313	+0.550	2795	+0.94	rank 7
19	23205	2020-02-13 → 2020-02-21	+33	+0.189	+0.389	2658	+0.71	rank 10
20	23205	2020-01-31 → 2020-02-13	+64	-0.080	-0.145	1657	+0.53	rank 17, SNAPHU unwrap
21	23205	2019-12-20 → 2020-01-31	+330	+0.222	+0.291	858	+0.35	rank 9, SNAPHU unwrap